# Minimal GitHub Guide

## Introduction

GitHub is a cloud-based platform that uses Git for version control, allowing you to store, track, and manage your projects efficiently. Whether you're building scripts, web applications, automation tools, or documentation, GitHub helps you maintain a complete history of your work and makes collaboration easier.

This guide focuses on the essential GitHub workflow for beginners:

- Configure Git on a new computer
- Connect GitHub using SSH authentication
- Copy an existing repository and make it your own
- Push your projects safely and efficiently

By completing these modules, you'll have a reliable GitHub setup that you can use for any future project.

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# Prerequisites

Before starting, ensure you have:

- A GitHub account
- Internet access
- Administrator access to your computer

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# Installation

**Linux (Ubuntu/Debian)**

```
sudo apt update
sudo apt install git -y
```

Verify:

```
git --version
```

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# Module 1: Initial Git Setup (One-Time)

Configure your Git identity.

```
git config --global user.name "Eugene Horfilla"
git config --global user.email "mreugene2005@gmail.com"
```

Verify your configuration:

```
git config --global --list
```

Expected output:

```
user.name=Eugene Horfilla
user.email=mreugene2005@gmail.com
```

---

# Module 2: GitHub SSH Setup (One-Time)

SSH allows Git to securely communicate with GitHub without repeatedly asking for credentials.

## Step 1: Generate an SSH Key

```
ssh-keygen -t ed25519 -C "mreugene2005@gmail.com"
```

Press **Enter** to accept the default file location.

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## Step 2: Start SSH Agent

```
eval "$(ssh-agent -s)"
```

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## Step 3: Add the SSH Key

```
ssh-add ~/.ssh/id_ed25519
```

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## Step 4: Display the Public Key

```
cat ~/.ssh/id_ed25519.pub
```

Copy the entire output.

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## Step 5: Add the SSH Key to GitHub

Navigate to:

```
GitHub
 └── Settings
     └── SSH and GPG Keys
         └── New SSH Key
```

Paste your public key and save.

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## Step 6: Test the Connection

```
ssh -T git@github.com
```

Expected result:

```
Hi <your-github-username>! You've successfully authenticated...
```

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# Module 3: Copy an Existing Repository and Make It Your Own

Use this when you want to take an existing GitHub project and create your own version.

## Step 1: Clone the Original Repository

```
git clone git@github.com:OTHER_USER/project.git
```

Example:

```
git clone git@github.com:dikoalam18/echo_linux101.git
```

Enter the project folder:

```
cd linux101
```

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## Step 2: Remove Original Git History

Remove the existing Git configuration:

```
rm -rf .git
```

This makes the project independent from the original repository.

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## Step 3: Initialize a New Git Repository

```
git init
```

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## Step 4: Stage All Files

```
git add .
```

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## Step 5: Create Your First Commit

```
git commit -m "Initial commit"
```

## Step 6: Create a New Repository on GitHub

Create a new empty repository in GitHub.

Example:

    linux101-copy

Important:

- Do NOT add a README
- Do NOT add a .gitignore
- Do NOT add a license

Leave the repository empty.

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## Step 7: Connect Your GitHub Repository

```
git remote add origin git@github.com:YOUR_USERNAME/linux101-copy.git
```

Example:

```
git remote add origin git@github.com:dikoalam18/linux101-copy.git
```

Verify:

```
git remote -v
```

Expected:

```
origin git@github.com:dikoalam18/linux101-copy.git (fetch)
origin git@github.com:dikoalam18/linux101-copy.git (push)
```

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## Step 8: Push to GitHub

```
git push -u origin main
```

Expected:

    Branch 'main' set up to track 'origin/main'

# Daily Git Commands

Check project status:

```
git status
```

Add changes:

```
git add .
```

Create a commit:

```
git commit -m "Describe changes"
```

Push changes:

```
git push
```

Download latest changes:

```
git pull
```

# Quick Reference

## New Computer Setup:

```
git config --global user.name "Eugene Horfilla"
git config --global user.email "mreugene2005@gmail.com"

ssh-keygen -t ed25519 -C "mreugene2005@gmail.com"
eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_ed25519"
cat ~/.ssh/id_ed25519.pub
```

Add the key to GitHub and test:

```
ssh -T git@github.com
```

**Copy Someone Else's Repository**

```
git clone git@github.com:OTHER_USER/project.git
cd project

rm -rf .git
git init

git add .
git commit -m "Initial commit"

git remote add origin git@github.com:YOUR_USERNAME/project.git
git push -u origin main
```

### Save Your Work

```
git add .
git commit -m "Updated project"
git push
```

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**Remember:** Git tracks changes locally, while GitHub stores those changes online. A typical workflow is:

Repeat these three commands whenever you want to save and publish your work. 🚀